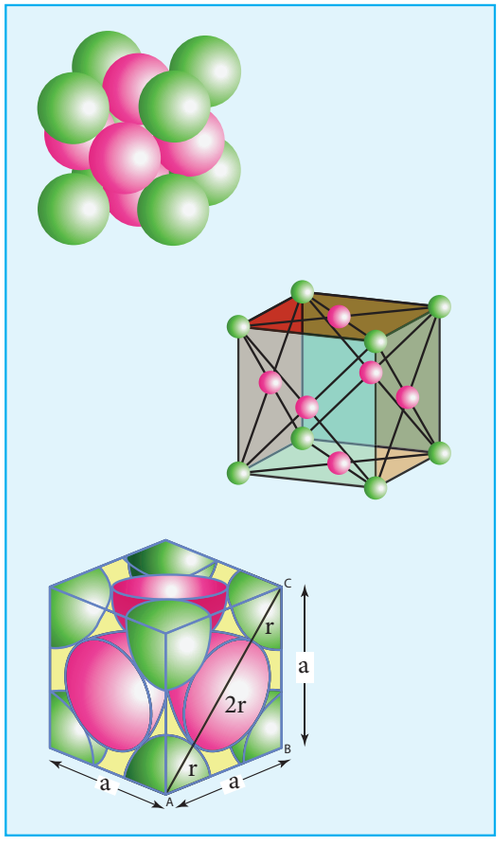
6.5.3 Face centered cubic unit cell.(FCC)



In a face centered cubic unit cell, identical atoms lie at each corner as well as in the centre of each face. Those atoms in the corners touch those in the faces but not each other.The coordination number is 2. The atoms in the face centre is being shared by two unit cells, each atom in the face centers makes $\left(\frac{1}{2}\right)$ contribution to the unit cell.

$$\therefore \text{ Number of atoms in a fcc unitcell } = \left(\frac{Nc}{8}\right)+\left(\frac{N_f}{2}\right)$$

$$=\left(\frac{8}{8}+\frac{6}{2}\right)$$

$$=(1+3)$$

$$=4$$

Drawing the crystal lattice on paper is not an easy task. The constituents in a unit cell touch each other and form a three dimensional network. This can be simplified by drawing crystal structure with the help of small circles (spheres) corresponding constituent particles and connecting neighbouring particles using a straight line as shown in the figure.

6.5.4 Calculations involving unit cell dimensions:

X-Ray diffraction analysis is the most powerful tool for the determination of crystal structure. The inter planar distance (d) between two successive planes of atoms can be calculated using the following equation form the X-Ray diffraction data

$$2d\sin\theta = n\lambda$$

The above equation is known as Bragg's equation.

Where

λ the wavelength of X-ray used for diffraction.

θ The angle of diffraction

By knowing the values of θ,λ and n we can calculate the value of d.

$$d = \frac{n\lambda}{2\sin\theta}$$

Using these values the edge of the unit cell can be calculated.

6.5.5 Calculation of density:

Using the edge length of a unit cell, we can calculate the density (ρ) of the crystal by considering a cubic unit cell as follows.

$$\text{Density of the unit cell } \rho = \frac{\text{mass of the unit cell}}{\text{volume of the unit cell}} \quad \ldots(1)$$